

Fact Sheet

CHINESE REUSABLE EXPERIMENTAL SPACECRAFT¹

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SUMMARY

China's "reusable experimental spacecraft" (Chongfu Shiyong Shiyan Hangtian Qi, or CSSHQ) is a technology demonstrator and experimental vehicle which is likely used for testing reusable space launch vehicle technologies, as well as on-orbit testing of new sensor technologies and satellite hardware for risk reduction. The CSSHQ has deployed a small satellite during each of its first three missions. It has demonstrated limited rendezvous and proximity operations with the satellites that it has released during its missions, but the spaceplane has near-zero feasibility as an orbital weapons system for attacking targets on the ground.²

BACKGROUND

- Similar to the United States' X-37B, China's CSSHQ is uncrewed and estimated to be relatively small.³
- The CSSHQ has historically been launched on top of China's Long March 2F rocket from Jiuquan Satellite Launch Center, and is thought to have concluded its missions with horizontal landings at the Lop Nur test site.
- China Aerospace Science and Technology Corporation (CASC) and China Aerospace Science and Industry Corporation (CASIC) are reported to be driving the development of China's experimental spaceplane program.⁴
- The U.S. military has catalogued CSSHQ test flights as occurring between 331 and 620 kilometers in altitude and 49.99° and 50.2° inclination orbit.⁵
- China has not specified which government entity is responsible for the spaceplane program.

Chinese Reusable Experimental Spacecraft Orbital Flight History

Launch Date	Launch Location	Landing Date	Landing Location	Time on Orbit	Vehicle
September 4, 2020	Jiuquan Satellite Launch Center	September 6, 2020	Lop Nur Test Site	2 days	Long March 2F
August 4, 2022	Jiuquan Satellite Launch Center	May 8, 2023	Lop Nur Test Site	276 days	Long March 2F
December 14, 2023	Jiuquan Satellite Launch Center	September 5, 2024	Lop Nur Test Site	268 days	Long March 2F
February 7, 2026	Jiuquan Satellite Launch Center	Still in orbit*	N/A*	N/A*	Long March 2F

* Information as of June 5, 2026.

DEBATE OVER MISSION AND RATIONALE FOR THE CHINESE REUSABLE EXPERIMENTAL SPACECRAFT

China's official objectives of the program include verification and space science experiments to "provide technical support for the peaceful use of space" and to "pave the way for more convenient and affordable round-trip methods for the peaceful use of space in the future."⁶ However, many have questioned and speculated upon its official mission.

Over the course of its four missions, the CSSHQ has tested multiple technologies, including vertical takeoff and horizontal landing, capture and docking operations, and rendezvous and formation flying. The first CSSHQ mission lasted from September 4 to September 6, 2020, during which time an unknown object was deployed and subsequently catalogued by the US military (Object A, 2020-063G).⁷ The object, which was believed to be a subsatellite, broadcast S-band transmissions for several weeks after the spaceplane's mission concluded.⁸ The second CSSHQ mission began on August 4, 2022 and concluded on May 8, 2023, totaling 276 days in orbit. On October 23, 2022, the CSSHQ raised its perigee⁹ and released a new object catalogued by the US military as OBJECT J (2022-093J).¹⁰ The CSSHQ is believed to have conducted rendezvous and proximity operations on multiple occasions, as well as at least two capture/docking operations with Object J in November-December 2022, January 2023, and February-March 2023. On at least five occasions, Object J maneuvered independently of the spaceplane, indicating that the subsatellite was equipped with independent propulsive capabilities.¹¹ The third CSSHQ mission began on December 14, 2023 and concluded on September 5, 2024, totaling 268 days in orbit. In late January 2024, CSSHQ raised its perigee, and on May 24, 2024, it released an unknown object (OBJECT G, 2023-195G) into orbit. After releasing Object G, CSSHQ performed a series of close maneuvers with the object and possible recapture operations¹² on June 5-7, June 11-12, June 14, and June 15-16, 2024.¹³ Following the third CSSHQ mission, Chinese press reporting emphasized the success of the experimental mission and the growing maturity of the spaceplane program.¹⁴ A fourth flight of the CSSHQ started in February 2026 with a launch from Jiuquan Satellite Launch Center via a Long March-2F (PRC Test Spacecraft 4, 2026-024A).¹⁵ The spaceplane was first catalogued to be in an orbit of 50° of 344 km x 590 km; about five days after its launch, it maneuvered to a circular 588 km x 597 km orbit.¹⁶ According to state news agency *Xinhua*, the spaceplane "will conduct technological verification for reusable spacecraft, providing technical support for the peaceful use of space."¹⁷

There is speculation that the CSSHQ could serve as a space weapon launch platform or a platform for surveillance, intelligence, and early-warning missions. China's *Global Times* outlet refuted these claims, reiterating that the system is "just a verification platform for testing the reentry glide return technology of a winged aircraft." The *Global Times* article was subsequently featured on the official website of the China Academy of Space Technology (CAST).¹⁸ As of May 2026, the CSSHQ has not approached nor maneuvered with non-Chinese space objects. Similar to the X-37B spaceplane, the CSSHQ's utility as a space-to-ground weapon is constrained by orbital mechanics. Any hyperkinetic weapon dropped from the spaceplane's bay would need to be equipped with its own thrusters capable of performing a deorbit burn. Based on the estimated size of the spaceplane, such a weapon would be unlikely to fit within the spaceplane's payload bay.¹⁹ However, co-orbital concerns remain around the CSSHQ's dual use capabilities, due to the spaceplane's demonstrated ability to release, dock, and maneuver with objects in low Earth orbit.

CHINESE REUSABLE EXPERIMENTAL SPACECRAFT FEATURES

Similar to the X-37B, the CSSHQ is uncrewed and appears to be relatively small.²⁰ China's use of the Long March 2F rocket for the first four flights suggests that the CSSHQ is comparatively larger in size than China's Shenlong spaceplane model released in 2000, due to the Long March 2F rocket's reported payload capacity of around 8 metric tons for low Earth orbit placement.²¹ The CSSHQ spaceplane is thought to return to Earth autonomously through a horizontal runway landing, similar to a conventional aircraft.

The CSSHQ's time spent in orbit increased significantly between its first flight, which lasted two days, and its second flight, which lasted 276 days. The time intervals between flights has decreased, from nearly two years between the first and second flight to seven months between the second and third flight.²²

MANEUVERABILITY OF THE CHINESE REUSABLE EXPERIMENTAL SPACECRAFT

During its second flight, the CSSHQ raised its orbit significantly, from 346 km by 593 km to 607 km by 597 km, rising to a nearly circular orbit. During the third CSSHQ flight, the spaceplane raised its orbit from 333 km by 348 km to 609 km by 601 km.²³ As stated above, the CSSHQ has demonstrated an ability to perform maneuvers and capture/docking operations with other objects equipped with independent propulsive systems. These maneuvers and orbit raises suggest that the CSSHQ possesses significant delta-v capability.

TRANSPARENCY CONCERNS

Much of the speculation over the Chinese Reusable Experimental Spacecraft stems from the program's secretive nature. China has not revealed many official details about the spaceplane's mission or on-orbit activities. Official statements are limited to launch announcements, which are released day-of, and applause for the program's success.²⁴ China has repeatedly claimed that the Chinese Reusable Experimental Spacecraft program is in line with the country's commitment to the "peaceful uses of outer space," but China has not registered the spaceplane nor its sub satellites with the United Nations for any of its past four missions, in contradiction to its legal responsibilities under the 1975 Registration Convention.²⁵

DEPLOYMENT OF SATELLITES

The Chinese Reusable Experimental Spacecraft has deployed satellites during each of the spaceplane's first three missions, although China has never announced or confirmed these deployments. Identification and announcements of deployed objects have been conducted by commercial SSA companies, and object cataloging has been done by the US military. The satellite deployed during the first mission demonstrated transmission broadcast capabilities,²⁶ and the satellite deployed during its second mission is thought to have displayed independent propulsive capabilities. The CSSHQ has conducted numerous close maneuvers and capture/docking operations with the deployed satellites.

ENDNOTES

1. The reason why the switch away from calling this “Shenlong” is that the Chinese government has never used that name for it; it comes from a photo taken nearly two decades ago of an object that had “Shenlong” painted on it. That object never went to space; its manufacturer has since moved on to different spaceplanes. There was nearly a decade between its flight and the flight of the Chinese Reusable Experimental Spacecraft, which has a different manufacturer. So while we cannot prove that the latter is not the Shenlong, we cannot prove it is either. So Chinese Reusable Experimental Spacecraft it is until we learn differently.
2. For more information about the Chinese reusable experimental spacecraft, please see: Victoria Samson and Kathleen Brett, editors, *Global Counterspace Capabilities: An Open Source Assessment*, Secure World Foundation, April 2026, <https://swfound.org/counterspace/>.
3. Daniel Shats and Peter Wood, “Chinese Spaceplane Programs,” *China Aerospace Studies Institute*, November 2021, <https://www.airuniversity.af.edu/Portals/10/CASI/documents/Research/Space/2021-12-06%20Spaceplanes.pdf>.
4. Shats and Wood, “Chinese Spaceplane Programs,” *ibid*.
5. Samson and Brett, *Global Counterspace Capabilities: An Open Source Assessment*, *ibid*.
6. “China successfully lands reusable experimental spacecraft,” *Xinhua*, September 6, 2024, <https://english.news.cn/20240906/651bb65c0c0b4dce896882d815c293a1/c.html>.
7. Data compiled from the public US military satellite catalog at <https://space-track.org>.
8. Andrew Jones, “China’s mystery spaceplane releases object into orbit,” *SpaceNews*, November 2, 2022, <https://spacenews.com/chinas-mystery-spaceplane-releases-object-into-orbit/>.
9. Andrew Jones, “China’s spaceplane raises orbit and national funding,” *SpaceNews*, October 25, 2022, <https://spacenews.com/chinas-spaceplane-raises-orbit-and-national-funding/>.
10. Data compiled from the public US military satellite catalog at <https://space-track.org>.
11. Andrew Jones, “China’s spaceplane conducted proximity and capture maneuvers with subsatellite, data suggests,” *SpaceNews*, May 11, 2023, <https://spacenews.com/chinas-spaceplane-conducted-proximity-and-capture-maneuvers-with-subsatellite-data-suggests/>.
12. Clayton Swope, Kari Bingen, Makena Young, and Kendra LaFave, editors, *Space Threat Assessment 2025*, *Center for Strategic and International Studies*, April 25, 2025, <https://www.csis.org/analysis/space-threat-assessment-2025>.
13. “The Chinese robotic Space Plane 3 and Object G: proximity operations [MULTIPLE UPDATES],” *SatTrackCam Leiden (b)log*, June 12, 2024, <https://sattrackcam.blogspot.com/2024/06/the-chinese-robotic-space-plane-3-and.html>.
14. “China successfully lands reusable experimental spacecraft,” *Xinhua*, September 6, 2024, <https://english.news.cn/20240906/651bb65c0c0b4dce896882d815c293a1/c.html>.
15. William Zheng, “China sends fourth ‘Shenlong’ reusable spacecraft mission into orbit,” *South China Morning Post*, February 7, 2026, <https://www.scmp.com/news/china/science/article/3342759/china-sends-fourth-shenlong-reusable-spacecraft-mission-orbit>.
16. Jonathan McDowell, “The Chinese spaceplane launched last night...” *Twitter*, February 7, 2026, <https://x.com/planet4589/status/2020169876250451979?>; Jonathan McDowell, “The Chinese spaceplane maneuvered...” *Twitter*, February 15, 2026, <https://x.com/planet4589/status/2023242545535283382?>.
17. “China successfully launches reusable experimental spacecraft,” *Xinhua*, February 7, 2026, <https://english.news.cn/20260207/396edb9335a94acc94b78443c5bd7f82/c.html>.
18. Shats and Wood, “Chinese Spaceplane Programs,” *ibid*.
19. Secure World Foundation, “X-37B Orbital Test Vehicle Factsheet,” May 2022, <https://swfound.org/media/207377/swf-x-37b-otv-fact-sheet-may-2022.pdf>.
20. Joanna Thompson, “China’s mysterious space plane released an unidentified ‘object’ in orbit, US intelligence reveals,” *Space.com*, May 27, 2023, <https://www.space.com/china-mysterious-space-plane-released-unidentified-object-in-orbit>.

21. Andrew Jones, "China's secretive spaceplane releases object into orbit," *SpaceNews*, May 26, 2024, <https://spacenews.com/chinas-secretive-spaceplane-releases-object-into-orbit/>.
22. Andrew Jones, "China's secretive spaceplane conducts proximity operations with small spacecraft," *SpaceNews*, June 13, 2024, <https://spacenews.com/chinas-secretive-spaceplane-conducts-proximity-operations-with-small-spacecraft/>.
23. Andrew Jones, "China's spaceplane raises orbit and national funding," *SpaceNews*, October 25, 2022, <https://spacenews.com/chinas-spaceplane-raises-orbit-and-national-funding/>.
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25. Samson and Brett, *Global Counterspace Capabilities: An Open Source Assessment*, ibid.
26. Andrew Jones, "China's mystery spaceplane releases object into orbit," *SpaceNews*, November 2, 2022, <https://spacenews.com/chinas-mystery-spaceplane-releases-object-into-orbit/>.
27. Andrew Jones, "China's spaceplane conducted proximity and capture maneuvers with subsatellite, data suggests," *SpaceNews*, May 11, 2023, <https://spacenews.com/chinas-spaceplane-conducted-proximity-and-capture-maneuvers-with-subsatellite-data-suggests/>.



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